Zomato Dataset Analysis

Project Title : Zomato Restaurant Data Analysis Using Python

Objective: The objective of this project is to analyze the Zomato restaurant dataset to identify trends in restaurant ratings, popular cuisines, restaurant locations, pricing patterns, and customer preferences. The analysis helps understand factors affecting restaurant performance and customer satisfaction.

Dataset Description :

The Zomato dataset contains information about restaurants such as restaurant name, location, cuisines served, ratings, votes received, average cost for two people, online ordering availability, and table booking options.

Dataset Features

- Restaurant Name
- Location
- Cuisines
- Rating
- Votes
- Average Cost for Two
- Online Order
- Table Booking
- Restaurant Type

Data Cleaning Steps :

The following preprocessing steps were performed:

1. Loaded the dataset using Pandas.

2. Removed duplicate records.

3. Checked for missing values and handled them appropriately.

4. Converted ratings into numeric format.

5. Cleaned the cost column by removing commas and converting it into integer values.

6. Removed unnecessary columns.

7. Standardized text values for consistency.

Exploratory Data Analysis (EDA)

1. Restaurant Ratings Distribution:

A histogram was created to understand the distribution of restaurant ratings.

Observation: Most restaurants have ratings between 3.5 and 4.5, indicating generally good customer satisfaction.

2. Top Cuisines

A bar chart was created to identify the most popular cuisines.

Observation: North Indian, Chinese, South Indian, and Fast Food are among the most frequently served cuisines.

3. Top Locations:

A count plot was used to determine restaurant concentration across locations.

Observation: Certain locations have a significantly higher number of restaurants, indicating major food hubs and business hotspots.

4. Cost vs Rating Scatter Plot:

A scatter plot was generated to examine the relationship between restaurant cost and ratings.

Observation: The relationship between cost and ratings is weak, suggesting that expensive restaurants do not necessarily receive better ratings.

Key Insights :

- Most restaurants maintain ratings between 3.5 and 4.5.
- North Indian and Chinese cuisines dominate the market.
- Restaurant businesses are concentrated in specific locations.
- Cost has a weak correlation with ratings.
- Restaurants with higher vote counts tend to have better ratings.

- Customer reviews and popularity significantly influence restaurant success.

Tools and Technologies Used :

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab / Jupyter Notebook
- GitHub

Methodology :

The project followed a structured data analysis approach:

1. Data Collection

2. Data Cleaning and Preprocessing

3. Exploratory Data Analysis (EDA)

4. Data Visualization

5. Insight Generation

6. Conclusion and Recommendations

Recommendations :

Based on the analysis, the following recommendations can be made:

- Restaurants should focus on maintaining food quality and customer service to improve ratings.
- Popular cuisines such as North Indian and Chinese should be prioritized based on customer demand.
- New restaurants can consider opening in high-demand locations to attract more customers.
- Restaurant owners should encourage customer reviews and ratings to increase visibility and trust.
- Pricing strategies should be designed carefully, as higher costs do not necessarily lead to better customer ratings.

Conclusion :

This analysis provided valuable insights into restaurant trends and customer preferences. Popular cuisines and strategic locations contribute significantly to restaurant visibility and success. The study also revealed that higher prices do not guarantee better customer ratings. These findings can help restaurant owners make data-driven decisions to improve customer satisfaction and business performance.

GitHub Repository Link :

https://git@github.com/Gangamma-yn/Zomato-Dataset-Analysis.git

Google Colab/Jupyter Notebook Link :

https://colab.research.google.com/drive/1Luv-_hCXgG72UUDj1lTypdbPd37LCmR9?usp=sharing

Screenshots of Outputs

	url	address	name	online_order	book_table	rate	votes	phone	location	rest_type	dish_liked	cuisines	approx_cost(for two people)	reviews_list	menu_item	listed_in(type)	listed_in(city)
0	https://www.zomato.com/bangalore/jalsa-banasha...	942, 21st Main Road, 2nd Stage, Banashankari, ...	Jalsa	Yes	Yes	4.1/5	775.0	080 42297555\r\n+91 9743772233	Banashankari	Casual Dining	Pasta, Lunch Buffet, Masala Papad, Paneer Laja...	North Indian, Mughlai, Chinese	800	[('Rated 4.0', 'RATED\n A beautiful place to ...	[]	Buffet	Banashankari
1	https://www.zomato.com/bangalore/spice-elephan...	2nd Floor, 80 Feet Road, Near Big Bazaar, 6th ...	Spice Elephant	Yes	No	4.1/5	787.0	080 41714161	Banashankari	Casual Dining	Momos, Lunch Buffet, Chocolate Nirvana, Thai G...	Chinese, North Indian, Thai	800	[('Rated 4.0', 'RATED\n Had been here for din...	[]	Buffet	Banashankari
2	https://www.zomato.com/SanchurroBangalore?cont...	1112, Next to KIMS Medical College, 17th Cross...	San Churro Cafe	Yes	No	3.8/5	918.0	+91 9663487993	Banashankari	Cafe, Casual Dining	Churros, Cannelloni, Minestrone Soup, Hot Choc...	Cafe, Mexican, Italian	800	[('Rated 3.0', "RATED\n Ambience is not that ...	[]	Buffet	Banashankari
3	https://www.zomato.com/bangalore/addhuri-udupi...	1st Floor, Annakuteera, 3rd Stage, Banashankar...	Addhuri Udupi Bhojana	No	No	3.7/5	88.0	+91 9620009302	Banashankari	Quick Bites	Masala Dosa	South Indian, North Indian	300	[('Rated 4.0', "RATED\n Great food and proper...	[]	Buffet	Banashankari
4	https://www.zomato.com/bangalore/grand-village...	10, 3rd Floor, Lakshmi Associates, Gandhi Baza...	Grand Village	No	No	3.8/5	166.0	+91 8026612447\r\n+91 9901210005	Basavanagudi	Casual Dining	Panipuri, Gol Gappe	North Indian, Rajasthani	600	[('Rated 4.0', 'RATED\n Very good restaurant ...	[]	Buffet	Banashankari

```
Data columns (total 17 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   url                          2650 non-null   object
 1   address                      2650 non-null   object
 2   name                         2650 non-null   object
 3   online_order                 2650 non-null   object
 4   book_table                   2650 non-null   object
 5   rate                         2270 non-null   object
 6   votes                        2650 non-null   int64
 7   phone                        2613 non-null   object
 8   location                     2649 non-null   object
 9   rest_type                    2637 non-null   object
 10  dish_liked                   1052 non-null   object
 11  cuisines                     2646 non-null   object
 12  approx_cost(for two people)  2647 non-null   object
 13  reviews_list                 2650 non-null   object
 14  menu_item                    2650 non-null   object
 15  listed_in(type)              2650 non-null   object
 16  listed_in(city)              2650 non-null   object
dtypes: int64(1), object(16)
memory usage: 352.1+ KB
```

	0
url	0
address	0
name	0
online_order	0
book_table	0
rate	380
votes	0
phone	37
location	1
rest_type	13
dish_liked	1598
cuisines	4
approx_cost(for two people)	3
reviews_list	0
menu_item	0

```python
df['rate'].unique()
```

```
array(['4.1/5', '3.8/5', '3.7/5', '3.6/5', '4.6/5', '4.0/5', '4.2/5',
       '3.9/5', '3.1/5', '3.0/5', '3.2/5', '3.3/5', '2.8/5', '4.4/5',
       '4.3/5', 'NEW', '2.9/5', '3.5/5', nan, '2.6/5', '3.8 /5', '3.4/5'
       '4.5/5', '2.5/5', '2.7/5', '4.7/5', '2.4/5', '2.2/5', '2.3/5',
       '3.4 /5', '-', '3.6 /5', '4.8/5', '3.9 /5', '4.2 /5', '4.0 /5',
       '4.1 /5', '3.7 /5', '3.1 /5', '2.9 /5', '3.3 /5', '2.8 /5',
       '3.5 /5', '2.7 /5', '2.5 /5', '3.2 /5', '2.6 /5', '4.5 /5',
       '4.3 /5', '4.4 /5', '4.9/5', '2.1/5', '2.0/5', '1.8/5', '4.6 /5',
       '4.9 /5'], dtype=object)
```

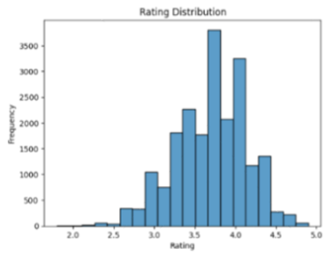
```python
print(X.isnull().sum())
print(y.isnull().sum())
```

```
url                              0
address                          0
name                             0
online_order                     0
book_table                       0
votes                            0
phone                            4
location                         0
rest_type                        0
dish_liked                     208
cuisines                         0
approx_cost(for two people)      0
reviews_list                     0
menu_item                        0
listed_in(type)                  0
listed_in(city)                  0
dtype: int64
35
```

```python
# Remove rows with missing values
df = df.dropna()

# Create features and target again
X = df[['online_order',
        'book_table',
        'votes',
        'approx_cost(for two people)']]

y = df['rate']
```

Explore relationships: cuisines vs rating,location hotspots,price vs rating using heatmaps,barcharts and wordclo

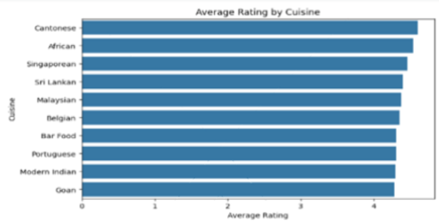
```python
sns.histplot(df['rate'],bins=20)
plt.title("Rating Distribution")
plt.xlabel("Rating")
plt.ylabel("Frequency")
plt.show()
```




```python
cuision_rating=df.groupby('cuisines')['rate'].mean().sort_values(ascending=False).head(10)
print(cuision_rating)
```

```
cuisines
Cantonese        4.600000
African          4.540000
Singaporean      4.460000
Sri Lankan       4.400000
Malaysian        4.372727
Belgian          4.350000
Bar Food         4.300000
Portuguese       4.300000
Modern Indian    4.294118
```

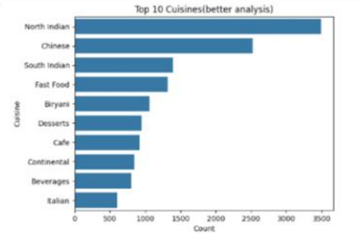
```python
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8,5))
sns.barplot(x=cuision_rating.values,y=cuision_rating.index)
plt.title("Average Rating by Cuisine")
plt.xlabel("Average Rating")
plt.ylabel("Cuisine")
plt.show()
```




```python
#split cuisines
cuisines_series=df['cuisines'].dropna().str.split(',')

#flatten list
all_cuisines=cuisines_series.explode()

#count
top_cuisines=all_cuisines.value_counts().head(10)
print(top_cuisines)
```

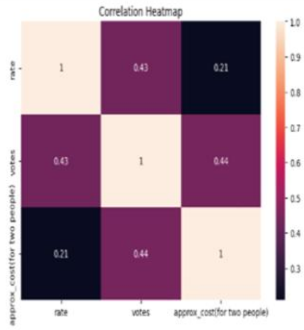
```
cuisines
North Indian    3494
Chinese         2521
South Indian    1393
Fast Food       1321
Biryani         1059
Desserts         946
Cafe             916
Continental      842
Beverages        799
Italian          604
Name: count, dtype: int64
```

```python
plt.title("Top 10 Cuisines(better analysis)")
plt.xlabel("Count")
plt.ylabel("Cuisine")
plt.show()
```




```python
plt.figure(figsize=(8,5))
sns.scatterplot(x=df['approx_cost(for two people)'],y=df['rate'])
plt.title("Cost vs Rating")
plt.xlabel("Cost for two")
plt.ylabel("Rating")
plt.show()
```

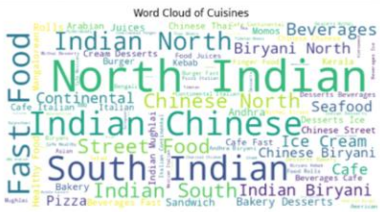

```python
corr = df[['rate','votes','approx_cost(for two people)']].corr()
plt.figure(figsize=(8,5))
sns.heatmap(corr,annot=True)
plt.title("Correlation Heatmap")
plt.show()
```




```python
import sys
!{sys.executable} -m pip install wordcloud
```

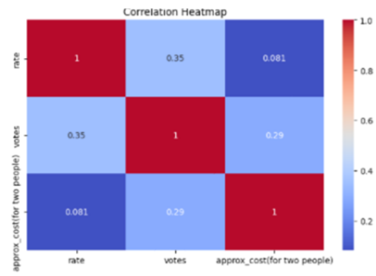
```
Requirement already satisfied: wordcloud in /usr/local/lib/python3.12/dist-packages (1.9.6)
Requirement already satisfied: numpy>=1.19.3 in /usr/local/lib/python3.12/dist-packages (from wordcloud) (2.0.2)
Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist-packages (from wordcloud) (11.3.0)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from wordcloud) (3.10.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (26.2)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib->wordcloud) (1.17.
```

```python
from wordcloud import WordCloud
text = ' '.join(df['cuisines'].dropna().astype(str))
wc=WordCloud(width=800,height=400,background_color='white').generate(text)
plt.figure(figsize=(8,5))
plt.imshow(wc,interpolation='bilinear')
plt.axis('off')
plt.title("Word Cloud of Cuisines")
plt.show()
```




```python
# Simplify cuisines (keep only first cuisine)
df['cuisines'] = df['cuisines'].fillna('Unknown')
df['cuisines'] = df['cuisines'].astype(str)
df['cuisines'] = df['cuisines'].apply(lambda x: x.split(',')[0])

# Reduce locations (keep top 10 locations only)
df['location'] = df['location'].fillna('Unknown')
top_locations = df['location'].value_counts().nlargest(10).index

df['location'] = df['location'].apply(
    lambda x: x if x in top_locations else 'other'
)

# Reduce dataset size safely
df = df.sample(n=min(len(df), 10000), random_state=42)

# Check result
print("Dataset Shape:", df.shape)
print(df[['cuisines', 'location']].head())
```

```
Dataset Shape: (182, 17)
         cuisines           location
21           Cafe       Banashankari
57          Asian       Banashankari
308  North Indian  Kumaraswamy Layout
205         Pizza       Banashankari
```



Train-Test Split

```python
x=df[['votes','approx_cost(for two people)','location','cuisines']]
y=df['rate']

from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(
    x,y,test_size=0.2,random_state=42)
```

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler,OneHotEncoder
from sklearn.pipeline import Pipeline
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error,r2_score
```

Preprocessing -

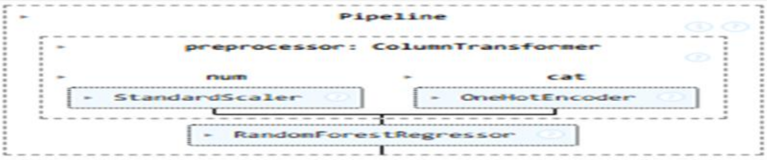
```python
from sklearn.compose import ColumnTransformer
numeric_features=['votes','approx_cost(for two people)']
categorical_features=['location','cuisines']
preprocessor=ColumnTransformer(
    transformers=[
        ('num',StandardScaler(),numeric_features),
        ('cat',OneHotEncoder(handle_unknown='ignore'),categorical_features)
    ]
```

```
# Remove '/5'
df['rate'] = df['rate'].str.replace('/5', '', regex=False)

# Replace invalid values with NaN
df['rate'] = df['rate'].replace(['NEW', '-', 'nan'], np.nan)

# Convert to float
df['rate'] = pd.to_numeric(df['rate'], errors='coerce')

# Remove rows with missing rate
df = df.dropna(subset=['rate'])
```

```
from sklearn.pipeline import Pipeline
from sklearn.ensemble import RandomForestRegressor

model = Pipeline([
    ('preprocessor' , preprocessor),
    ('regressor', RandomForestRegressor(
        n_estimators=50,
        max_depth=10,
        random_state=42
    ))
])
```

train model

```
# Train the model
model.fit(x_train, y_train)
```




```
# Make predictions
y_pred = model.predict(x_test)
```

Evaluation

```
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)

print("Root Mean Squared Error:", rmse)
print("R-squared:", r2)
```

```
Root Mean Squared Error: 0.35013506492882984
R-squared: -0.1314005507302689
```

Executive Summary :

This project analyzed the Zomato restaurant dataset using Python, Pandas, Matplotlib, and Seaborn. The objective was to identify trends in restaurant ratings, cuisines, locations, and pricing. Data cleaning and preprocessing were performed before conducting exploratory data analysis. Visualizations revealed that most restaurants receive ratings between 3.5 and 4.5. North Indian and Chinese cuisines emerged as the most popular food categories. Certain locations showed a higher concentration of restaurants, indicating important business hubs. The analysis also found only a weak relationship between restaurant cost and customer ratings. Overall, the project demonstrates how data analytics can provide actionable insights into customer preferences and restaurant performance.